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Check your grasp!

What will be the quanta requirement for complete light reaction which releases 6 oxygen molecules?

Solution: Complete light reaction releases 6 oxygen molecules. If one molecule of oxygen evolution requires 8 quanta means, for 6 oxygen molecules 6 × 8 = 48 quanta of light required for complete light reaction.

13.13 Dark Reaction or C_3 Cycle or Biosynthetic Phase or Photosynthetic Carbon Reduction (PCR)Cycle

Biosynthetic phase of photosynthesis utilises assimilatory powers(ATP and NADPH + H^+) produced during light reaction are used to fix and reduce carbon dioxide into carbohydrates. This reaction does not require light. Therefore, it is named Dark reaction. Ribulose 1,5 bisphosphate (RUBP) act as acceptor molecule of carbon dioxide and fix the CO_2 by RUBISCO enzyme. The first product of the pathway is a 3- carbon compound (Phospho Glyceric Acid) and so it is also called as C_3 Cycle. It takes place in the stroma of the chloroplast. **M. Melvin Calvin, A.A. Benson** and their co-workers in the year 1957 found this path way of carbon fixation. Melvin Calvin was awarded Nobel Prize for this in 1961 and this pathway named after the discoverers as **Calvin-Benson** Cycle. Dark reaction is temperature dependent and so it is also called thermo-chemical reaction.

Dark reaction consists of three phases: (Figure 13.18).

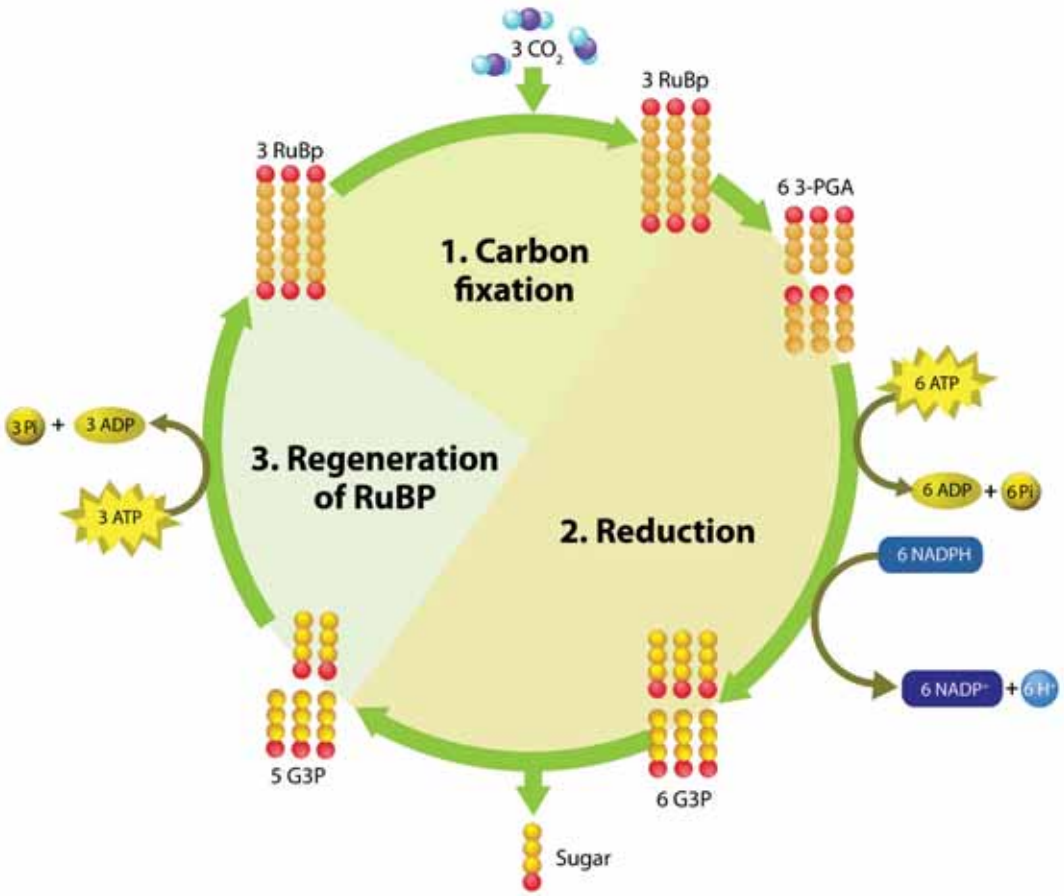


Figure 13.18: Phases of Calvin Cycle